

Joint inversion of seismic traveltime and frequency-domain airborne electromagnetic data for hydrocarbon exploration

Jide Nosakare Ogunbo¹, Guy Marquis², Jie Zhang³, and Weizhong Wang⁴

ABSTRACT

Geophysical joint inversion requires the setting of a few parameters for optimum performance of the process. However, there are yet no known detailed procedures for selecting the various parameters for performing the joint inversion. Previous works on the joint inversion of electromagnetic (EM) and seismic data have reported parameter applications for data sets acquired from the same dimensional geometry (either in two dimensions or three dimensions) and few on variant geometry. But none has discussed the parameter selections for the joint inversion of methods from variant geometry (for example, a 2D seismic travel and pseudo-2D frequency-domain EM data). With the advantage of affordable computational

cost and the sufficient approximation of a 1D EM model in a horizontally layered sedimentary environment, we are able to set optimum joint inversion parameters to perform structurally constrained joint 2D seismic traveltime and pseudo-2D EM data for hydrocarbon exploration. From the synthetic experiments, even in the presence of noise, we are able to prescribe the rules for optimum parameter setting for the joint inversion, including the choice of initial model and the cross-gradient weighting. We apply these rules on field data to reconstruct a more reliable subsurface velocity model than the one obtained by the traveltime inversions alone. We expect that this approach will be useful for performing joint inversion of the seismic traveltime and frequency-domain EM data for the production of hydrocarbon.

INTRODUCTION

The seismic method is by far the most widespread geophysical technique used for hydrocarbon exploration at depth from hundreds to thousands of meters. However, near-surface (here defined as the top 200–300 m of the subsurface) seismic imaging can be very challenging due to near-surface heterogeneity: rugged topography, large velocity variations, or hidden low-velocity layers (Zhang, 2009). The near-surface effects are observed through the observed time shifts from the linear and hyperbolic moveout of seismic refraction and reflection events, respectively (Yilmaz, 2001). Existing seismic techniques for solving near-surface problems include delay-time solution (Gardner, 1967), generalized linear inversion (Hampson and Russell, 1984), first-arrival traveltime tomography (Zhang and Toksöz, 1998; Zhu et al., 2008), refraction traveltime migration (Zhang, 1997), refraction wavefield migration (Rockwell, 1967; Hill,

1987; Zhang, 2006), early-arrival waveform tomography (Sheng et al., 2006), and surface-wave dispersion curve inversion (Xia et al., 1999; Ernst, 2007). For many seismic data processing problems, accurate near-surface imaging is a key for producing reliable deep subsurface images. Indeed, as the source and receivers are located at the earth's surface, the effects of the near-surface heterogeneity must be corrected before proper subsurface imaging. For example, long-wavelength statics corrections are often calculated from a near-surface velocity model derived from any of the near-surface techniques listed above (Hatherly et al., 1994). Effective short-wavelength statics corrections are subsequently calculated using refraction (Colombo et al., 2016) and/or reflection (Wiggins et al., 1976; Taner and Koehler, 1981) data and then applied to reflection data.

Nonuniqueness is inherent to the inversion of geophysical data (Zhdanov, 2015). One way it may however be addressed is by considering multiple subsurface physical properties (Gallardo and

Manuscript received by the Editor 25 February 2017; revised manuscript received 2 November 2017; published ahead of production 07 December 2017; published online 08 February 2018.

¹University of Science and Technology of China, School of Earth and Space Sciences, Hefei, China and Earth Resources Laboratory, MIT, Cambridge, Massachusetts, USA. E-mail: nosajide@yahoo.com.

²Formerly Shell International Exploration and Production, Houston, Texas, USA; presently Institut de Physique du Globe de Strasbourg; UMR 7516, Université de Strasbourg/EOST, CNRS; 5 rue René Descartes, Strasbourg, France. E-mail: guy.marquis.gp@gmail.com.

³University of Science and Technology of China, School of Earth and Space Sciences, Hefei, China. E-mail: jzhang25@ustc.edu.cn.

⁴GeoTomo LLC, Houston, Texas, USA. E-mail: weizhong.wang@geotomo.com.

© 2018 Society of Exploration Geophysicists. All rights reserved.

Meju, 2004; Abubakar et al., 2012). Combining two or more independent geophysical imaging methods can provide extra information that can reduce the degree of nonuniqueness of subsurface characterization. In particular, seismic and electromagnetic (EM) methods can be complementary for near-surface imaging: For example, low-velocity zones that are impossible to image by the seismic refraction method may coincide with an electrical conductivity anomaly (Zhang and Morgan, 1996).

EM methods have been successful in near-surface and deep geophysical surveys for mineral (Smith, 2014), hydrocarbon (Eidemo et al., 2002; Constable and Srnka, 2007; Hesthammer et al., 2012), geothermal (Muñoz, 2014) and groundwater exploration (Auken et al., 2003), permafrost delineation (Minsley et al., 2012), and deep lithospheric sounding (Meqbel et al., 2014). However, EM methods may not necessarily be sensitive to stratigraphic structures that can be well-resolved by seismic methods. In comparison with ground-based EM methods, airborne EM methods can quickly and effectively acquire large quantities of data in any type of environment with excellent positioning accuracy (Vallée et al., 2011). Although 3D EM inversion is still very costly in computation (Schamper et al., 2012), 1D EM model interpretation is often a sufficient approximation in a horizontally layered sedimentary environment in which sensitivity to lateral changes in conductivity is negligible (Spies and Frischknecht, 1991).

Joint inversion of seismic and EM data has been the subject of extensive industry research over the past few years, with a strong effort on combining seismic and marine controlled-source electromagnetic (mCSEM) data for hydrocarbon exploration. Indeed, because the reservoir resistivity obtained from mCSEM has been proposed as a direct hydrocarbon indicator, some of the early joint inversion work focused on relating mCSEM with other potential hydrocarbon indicators, such as amplitude variation with offset/amplitude variation with angle (Hoversten et al., 2006; Lien and Mannseth, 2011; Chen and Hoversten, 2012). Other groups have focused on using seismic as a structural constraint for the resistivity inversion (Hu et al., 2009; Brown et al., 2012). Abubakar et al. (2012) apply petrophysical and structural constraints for the joint inversion of controlled-source electromagnetic (CSEM) data with surface seismic full-waveform data. Although these results are not directly addressing the near-surface imaging problems that are the focus of this paper, they have played an important role in establishing workflows and concepts that can be applied to a wide range of problems.

In parallel to the above efforts, several industry groups have looked at the integration of seismic and nonseismic data to specifically address near-surface seismic velocity problems. Colombo and Keho (2010) perform structurally constrained joint nonseismic (land EM and gravity) and seismic inversion to solve near-surface problems in Saudi Arabia. Colombo et al. (2012, 2015, 2016) enforce structural constraints to perform joint inversion of high-resolution EM, gravity, and seismic data sets. Pineda et al. (2015) use results from frequency-domain EM, ground-penetrating radar, and DC resistivity surveys to improve velocity models in complex near-surface environments in Algeria.

Some classes of constraints that are applied in joint geophysical inversions include, using the parametric relationship between physical properties (Jegen et al., 2009), applying compositional relationships (Lelièvre et al., 2012), or petrophysical (Hoversten et al., 2006; Abubakar et al., 2012) relationships, and applying structural constraints (Gallardo and Meju, 2004). Moorkamp (2017) gives a

review of the different approaches for joint interpretation (including joint inversion) of EM with other geophysical data.

There are many ways that joint inversion can be implemented. One can invert for a single physical property from multiple data sets, e.g., DC resistivity and magnetotellurics to image resistivity (Vozoff and Jupp, 1975); one could use multiple approaches on a single data set, e.g., joint refraction traveltimes migration and tomography (Zhang and Toksöz, 1998); or, and this is the case for this paper, one can invert independent geophysical data sets for several physical properties, e.g., inverting jointly seismic refraction, gravity, and magnetotellurics (Moorkamp et al., 2011).

At the risk of oversimplification, approaches for inverting together independent geophysical data sets, each sensitive to a different physical property, can be broadly divided into two main categories: petrophysically based and structurally based. Petrophysically based approaches relate the different physical properties through some predefined or adaptive relationships, whereas structurally based approaches focus on having the changes in physical properties occur at similar locations. In practice, these approaches are often combined (Abubakar et al., 2012).

Gallardo and Meju, (2003, 2004) describe how to apply structural constraints by cross gradient in a joint inversion. They successfully applied their concept on the joint inversion of 2D DC resistivity and seismic traveltimes data. Moorkamp et al. (2011) investigate the comparison between imposing direct functional between seismic velocity and conductivity and structural constraints. They conclude that the cross-gradient (structural) approach should be the first-choice constraint for regions where a direct relationship between physical parameters is uncertain.

Gao et al. (2010, 2012) have proposed to link reservoir seismic and EM properties through velocity-resistivity relationships. Sun and Li (2012) and Carter-McAuslan et al. (2015) detail how fuzzy *c*-means clustering can be used for coupling physical properties in joint inversion.

The work presented in this paper focuses on improving near-surface velocity models in hydrocarbon exploration workflows, i.e., in mostly sedimentary environments. We only very rarely have access to near-surface petrophysical information, and therefore we follow the conclusions of Moorkamp et al. (2011) and opt for a cross-gradient, i.e. structural, approach. Due to the specific issues inherent with airborne EM data, we made several adaptations to the methods presented above.

In the references given above, the joint inversions have been performed on the same grids for the complementary models used; for instance, Gallardo and Meju (2003, 2004) use the same grids for the 2D velocity and resistivity models in the joint inversion. In this study, however, we propose to implement the joint inversion of seismic and frequency-domain airborne EM data from two different grids: a 2D grid for the velocity model and a pseudo-2D grid (multiple 1D resistivity models stitched together because of the affordable cost of 1D EM interpretation over 3D EM inversion) for the resistivity model, imposing structural similarity constraints. These constraints are enforced by defining a cross-gradient function that depends on the direction of the variations of physical properties (in this case, velocity and resistivity values) rather than the magnitude of changes (Gallardo and Meju, 2003). In most realistic field cases, the number of cells in the 2D grid for the velocity model is smaller than that of the resistivity model. Because cross-gradient computation requires using collocated medium parameters, using

the same number of cells in the grid, the pseudo-2D resistivity model is interpolated to equate the size of the velocity model. Our approach has some similarity to that proposed by Garofalo et al. (2015) for the joint inversion of surface-wave dispersion with DC resistivity and refracted data, except that we do not extract local 1D models to relate the different parameters: Instead, we take advantage of the dense sampling of airborne EM data to do this in 2D.

In the following sections, we will first briefly discuss the 2D seismic traveltime inversion method by Zhang and Toksöz (1998), a pseudo-2D frequency-domain airborne EM inversion method through regularization, and their integration into a structurally constrained joint inversion. We will demonstrate the application of joint inversion methodology, first to a synthetic data set and then to field data from coincident seismic and frequency-domain airborne EM surveys in Canada.

SEISMIC FIRST-ARRIVAL TRAVELTIME TOMOGRAPHY

Seismic first-arrival traveltime tomography has been extensively applied in near-surface imaging in oil and gas exploration industry. Recent efforts are focused on the seismic wavefront traveltime inversions. The wavefront traveltimes represent the first arrivals from a point source to the entire model. The calculation of the wavefront traveltime and raypath in a grid model can be conducted by methods such as an eikonal solver (Vidale, 1988) or graph method (Moser, 1991). The finite-difference eikonal solver involves the calculation of traveltimes first, and then inferring seismic raypaths back from receivers to the sources by analyzing the maximum gradient of the wavefront traveltimes. However, a graph method calculates traveltimes and raypaths at the same time. A higher order graph method by using more local rays in a template can not only be more accurate, but also more time consuming. The advantage of the original eikonal solver is its fast calculation speed, but it may not be accurate for models with complex velocity variations; therefore, many improvements have been developed with the cost of the additional computation efforts (van Trier and Symes, 1991; Qin et al., 1992; Hole and Zelt, 1995; Sethian and Popovici, 1999). Lelièvre et al., (2011a, 2011b) implement the fast marching method on unstructured 2D triangular grids and 3D tetrahedral grids for efficient generation of complicated subsurface geometries found in hard-rock environments. Their inversion approach requires no ray tracing and uses sensitivity values computed directly from the forward solution.

We use a rectilinear (rectangular) mesh in this study because we are considering simple structures from sedimentary environments and our forward code results are within acceptable accuracies. After comparing the performance of eikonal solvers and graph methods with the tradeoffs between accuracy and efficiency, we improve the graph method by Zhang and Toksöz (1998) by implementing a graph template up to 70th order (Zhang, 1997). A bucket algorithm is also applied to speed up the sorting process in calculation by several orders of computational time (Klimes and Kvasnicka, 1993). For the first-arrival traveltime inversion, we solve a Gauss-Newton inverse problem by minimizing an objective function that includes a data misfit term and a model Tikhonov regularization term. We apply the same regularization approach to frequency-domain airborne EM data for inversion so that the two methods can be easily integrated for joint inversion. From the previous publications and our numerical experiments, we find that the traveltime tomography can

easily converge from any starting model quickly because of its efficient implementation, which calculates its true derivative (raypath) associated with the model iteratively (Zhang, 1997; Zhang and Toksöz, 1998). However, inverting traveltimes is highly nonunique. With an effective regularization approach applied, the inversion may lead to a single high-probability solution but in low resolution sometimes. That suggests seismic traveltime data alone may not offer sufficient information to constrain the subsurface geology. Therefore, integrating multiple data sets is essential to improve geophysical imaging.

EM MODELING AND INVERSION THEORY

A frequency-domain airborne EM survey is based on the electromagnetic induction principle whereby the subsurface is energized by an alternating current in a transmitter (Nabighian and Macnae, 1991). The alternating current induces a primary magnetic field. This primary field will in turn induce Eddy currents in subsurface conductors, and these secondary currents in turn produce a secondary magnetic field (Nabighian and Macnae, 1991; Palacky and West, 1991). During frequency-domain EM data acquisition, the primary and secondary fields are recorded by the receiver. Frequency-domain EM data are traditionally presented as real (in-phase) and imaginary (quadrature) components of the ratio of secondary to primary field (Palacky and West, 1991). Because the magnitude of the secondary field is much smaller than that of the primary field (Nabighian and Macnae, 1991), their ratio is usually given in parts per million (ppm).

Acquisition and 1D forward modeling of frequency-domain airborne EM field

For frequency-domain airborne EM data acquisition, the airborne EM system typically carries several transmitter-receiver coil pairs, which can be coplanar, coaxial, or a combination. This allows multi-component data measurements of the magnetic field as horizontal coplanar (HCP) pairs produce and measure the vertical components. In addition, vertical coaxial pairs produce and measure the horizontal components of the magnetic field (Zhdanov, 2009).

For the cases considered here in which the transmitter is a horizontal loop, the transverse electric field (TE) mode will be used to compute the transient voltage measured over a horizontally layered, isotropic earth. The TE mode is generated by a vertical magnetic dipole (VMD) source, and its magnetic field is transverse to an axis of reference, z -axis in a 1D layered earth. Assuming an airborne EM system with an HCP transmitter and receiver pair, we can compute the TE-mode vertical magnetic field at any depth z . Material properties, μ_n , ϵ_n and σ_n are defined in each layer of thickness h_n , where μ_n is the magnetic permeability (H/m), ϵ_n is the dielectric permittivity in (F/m), and σ_n is electrical conductivity (S/m) of layer n . In this work, we assume that ϵ_n and μ_n are everywhere equal to the permittivity and permeability of free space, respectively; i.e., $\epsilon_n = \epsilon_0$ and $\mu_n = \mu_0$. The depth of interface n is denoted as z_n .

In cylindrical coordinates, the vertical magnetic field at angular frequency ω , produced by a dipole source and measured at a receiver located at altitudes h and z , respectively, above the earth surface and separated by a horizontal distance r is given as (Ward and Hohmann, 1988):

$$H_z(\omega) = \frac{m}{4\pi} \int_0^\infty (e^{-u_0(z+h)} + r_{TE} e^{u_0(z-h)}) \frac{\gamma^3}{u_0} J_0(\gamma r) d\gamma, \quad (1)$$

where $m = IS$ is the VMD moment, I is the low-frequency alternating current, S is the surface area of the transmitter loop, γ is an integration variable, r_{TE} is the TE-model reflection coefficient, and u is related to the classical EM wave number. The Hankel transform in equation 1 is solved by convolution filter (Anderson, 1979; Ward and Hohmann, 1988).

Pseudo-2D frequency-domain airborne EM inversion

A 1D interpretation of EM data can be misleading for complex earth models (Patra and Mallick, 1980; Spies and Frischknecht, 1991). On the other hand, the computational overhead of true 2D and 3D frequency-domain airborne EM data imaging is significant (Schultz and Ruppel, 2005; Schamper et al., 2012). A laterally constrained inversion approach is used by Auken and Christiansen (2004) for resistivity inversion. Schultz and Ruppel (2005) also perform a pseudo-2D inversion of multiple 1D EM data sets by applying 1D forward modeling and the model regularization in 2D. Schamper et al. (2012) use the same idea for the inversion of multifrequency and multicomponent ground-based electromagnetic induction data. In this study, we solve a pseudo-2D frequency-domain airborne EM nonlinear inversion problem following the same approach for high computation efficiency. Over a 2D resistivity model $m_{j,i}$, the field data acquired at multiple frequencies d_i at position i form pseudo-2D field data. Similarly, the forward responses $[G_i(m_{j,i})]$ computed at multiple frequencies at positions i form a pseudo-2D computed data set. Each $[G_i(m_{j,i})]$ is computed using 1D model parameters $m_{j,i}$ directly below i . The acquired pseudo-2D data vector is defined as $\mathbf{d}_e = [d_1, d_2, d_3, \dots, d_i]$ whereas the computed pseudo-2D data vector is $\mathbf{G}_e(\mathbf{m}_e) = [G_1(m_{j,i}), G_2(m_{j,i}), G_3(m_{j,i}), \dots, G_i(m_{j,i})]$ where $\mathbf{m}_e = m_{j,i}$.

Giving $\mathbf{m}_e^k$ as the model parameter at k th iteration, $\Delta\mathbf{m}_e^{k+1}$ as the model update at k th iteration, the objective function can be formulated as a regularized least-squares inverse problem:

$$\phi(\mathbf{m}_e) = \|\mathbf{W}_e(\mathbf{m}_e - \mathbf{G}_e(\mathbf{m}_e))\|^2 + \tau_e \|\mathbf{L}\mathbf{m}_e\|^2, \quad (2a)$$

where $\mathbf{L}$ is the regularization operator (e.g., the Laplacian operator) and τ_e is the smoothing tradeoff parameter. For continuous and differentiable $\mathbf{G}_e(\mathbf{m}_e)$, Taylor's expansion about an initial model gives a linearized approximation for $\mathbf{G}_e(\mathbf{m}_e) = \mathbf{A}_e \Delta\mathbf{m}$, where $\mathbf{A}_e$ is the sensitivity matrix whose forward finite-difference approximation, on a linear scale, is given as

$$\mathbf{A}_{e(i,j)} = \frac{G_i(m_{j,i} + \delta) - G_i(m_{j,i})}{\delta}, \quad (2b)$$

where δ is the perturbation factor, which is determined empirically (Meju, 1994). Farquharson and Oldenburg (1996) and Farquharson et al. (2003) use an approximate adjoint field that integrates the scalar product of an adjoint electric field, to obtain sensitivity matrix. Although this approach is reported to be fast (Farquharson et al., 2003), we prefer to use the logarithmically scaled version (Zhdanov, 2015) of the approximation in equation 2b due to the ease of implementation. Therefore, substituting the linearized value of

$\mathbf{G}_e(\mathbf{m}_e)$ into equation 2a, differentiating it, and equating the derivative to zero will give

$$\begin{aligned} & ((\mathbf{W}_e \mathbf{A}_e)^T (\mathbf{W}_e \mathbf{A}_e) + \tau_e \mathbf{L}^T \mathbf{L} + \epsilon_e^k \mathbf{I}) \Delta\mathbf{m}_e^{k+1} \\ &= (\mathbf{W}_e \mathbf{A}_e)^T \mathbf{W}_e \Delta\mathbf{d}_e - \tau_e \mathbf{L}^T \mathbf{L} \mathbf{m}_e^k, \end{aligned} \quad (3)$$

where $\Delta\mathbf{d}_e$ is the data difference vector, $\mathbf{W}_e$ is the EM data weight, and $\epsilon_e^k \mathbf{I}$ is an iteration variable damping term (Zhang and Toksöz, 1998). The variable ϵ_e^k is added to control the influence of the small eigenvalues and it is defined as $\epsilon_e^k = \alpha \times \text{rhs}$, where α is an empirical parameter (approximately 0.1) and rhs is the root-mean square (rms) misfit norm of the right-hand side of equation 3. For a large rhs, which indicates that the misfit is still large, and thus the objective function is still highly nonlinear, a large ϵ_e^k is applied to generate minimal model updates. However, for a small rhs, small ϵ_e^k allows more whole model updates, thus, speeding up the convergence. Equation 3 can be inverted to find $\Delta\mathbf{m}_e^{k+1}$ using a conjugate gradient (CG) inversion method. Therefore, the model parameter at $(k+1)$ th iteration is then calculated as

$$\mathbf{m}_e^{k+1} = \mathbf{m}_e^k + \Delta\mathbf{m}_e^{k+1}, \quad k = 1, 2, 3, \dots, N. \quad (4)$$

The pseudo-2D frequency-domain airborne EM model is made possible by the use of a 2D regularization (Laplacian) operator $\mathbf{L}$. This eventually results in a pseudo-2D sensitivity matrix, $\mathbf{A}_e$. Because $\mathbf{L}$ can exist in any dimension (one dimension, two dimensions), it is thus trivial to extend this concept to pseudo-3D from either 1D or 2D combinations.

STRUCTURALLY CONSTRAINED JOINT INVERSION

A 2D subsurface structure can be characterized by seismic velocity and resistivity. Although these physical properties may not have direct relationship between them (Zhang and Morgan, 1996), their structural similarity can be constrained by the minimization of the cross product of their model gradients (i.e., the cross gradient), which depends on the direction of the variations rather than the magnitude of the changes in properties (Gallardo and Meju, 2003).

Because the structural similarity is a measure that is only direction dependent, rather than the magnitude, mathematically the cross-gradient value of two models (for example, resistivity and velocity) of zero will indicate that the models are structurally similar. For example, the cross-gradient method is zero valued for the following cases, when

- 1) both properties have zero gradients, for example, in background or homogeneous resistivity and velocity layers
- 2) both properties with gradients in the same direction, for example, in geology where resistivity and velocity models have positive correlation; i.e., an increase in velocity corresponds to a high-resistive layer and vice versa
- 3) both properties have gradients in opposite directions, for example, where the resistivity and velocity models have negative correlation; consider a high-velocity, low-resistive layer
- 4) one property with nonzero gradient and the second property has zero gradient, for example, the boundary between fresh and salt

groundwater constituting a strong electrical resistivity boundary with no change in seismic velocity (Gallardo, 2004).

Defining the cross gradient t , as a structural constraint (Gallardo and Meju, 2003, 2004), we impose the joint inversion objective function as follows:

$$\phi(\mathbf{m}_e, \mathbf{m}_s) = (\|\mathbf{W}_e(\mathbf{d}_e - \mathbf{G}_e(\mathbf{m}_e))\|^2 + \tau_e \|\mathbf{Lm}_e\|^2) + (\|\mathbf{W}_s(\mathbf{d}_s - \mathbf{G}_s(\mathbf{m}_s))\|^2 + \tau_s \|\mathbf{Lm}_s\|^2) + \lambda \|t\|^2. \quad (5)$$

All parameters retain their previous definitions as for frequency-domain airborne EM above, but the ones with subscript s correspond to the seismic method, where $\mathbf{d}_e$ is the pseudo-2D frequency-domain airborne EM data $\mathbf{d}_s$ is the observed first-arrival time data, and $\mathbf{G}_s(\mathbf{m}_s)$ is the computed first-arrival time data over a 2D slowness model ($\mathbf{m}_s$) using the improved 2D traveltimes code of Zhang and Toksöz (1998). Equation 5 defines the usual objective function to be minimized with the first two terms on the right side corresponding to data misfits with a regularization term, and the third term is the coupling term between two or more models that are jointly inverted (compare with the equations in Gallardo and Meju, 2003, 2004; Hu et al., 2009; Zhdanov, 2009, 2015; Gao et al., 2010; Moorkamp, 2017). The variables $\mathbf{W}_e$ and $\mathbf{W}_s$ are the EM and seismic data weights, respectively; these correspond to the inverse of covariance matrix as presented in Gallardo and Meju (2003, 2004) and Hoversten et al. (2006). The value λ is the cross-gradient weight. The selection of the optimum λ is discussed later. The cross-gradient constraint t is given as (Gallardo and Meju, 2003, 2004)

$$t(\log(\mathbf{m}_e), \mathbf{m}_s) = \nabla \log(\mathbf{m}_e(x, z)) \times \nabla \mathbf{m}_s(x, z), \quad (6)$$

where $\mathbf{m}_e$ and $\mathbf{m}_s$ are the resistivity and slowness models, respectively. It should be noted that the cross product, in equation 6, is inherently zero for 1D geometry because the physical properties both vary only in the vertical direction. Thus, a cross-gradient constraint cannot be imposed while jointly inverting the 1D velocity and resistivity structures. Structural constraints have to be introduced in a different way for these 1D problems (Moorkamp et al., 2010). Details of the evaluation of equation 6 are given by Gallardo and Meju (2003). After substituting equation 6 into equation 5 and equating its derivative with respect to the model parameters to zero, the objective function can be minimized, giving

$$\begin{pmatrix} ((\mathbf{W}_e \mathbf{A}_e)^T \mathbf{W}_e \mathbf{A}_e + \tau_e \mathbf{L}^T \mathbf{L}) + \lambda \mathbf{B}_e^T \mathbf{B}_e & \lambda \mathbf{B}_e^T \mathbf{B}_s \\ \lambda \mathbf{B}_s^T \mathbf{B}_e & ((\mathbf{W}_s \mathbf{A}_s)^T \mathbf{W}_s \mathbf{A}_s + \tau_s \mathbf{L}^T \mathbf{L}) + \lambda \mathbf{B}_s^T \mathbf{B}_s \end{pmatrix} \times \begin{pmatrix} \Delta \mathbf{m}_e^{k+1} \\ \Delta \mathbf{m}_s^{k+1} \end{pmatrix} = \begin{pmatrix} \beta((\mathbf{W}_e \mathbf{A}_e)^T \mathbf{W}_e \Delta \mathbf{d}_e - \tau_e \mathbf{L}^T \mathbf{Lm}_e^k) - \lambda \mathbf{B}_e^T t \\ ((\mathbf{W}_s \mathbf{A}_s)^T \mathbf{W}_s \Delta \mathbf{d}_s - \tau_s \mathbf{L}^T \mathbf{Lm}_s^k) - \lambda \mathbf{B}_s^T t \end{pmatrix}. \quad (7)$$

The parameters $\mathbf{B}_e$ and $\mathbf{B}_s$ are the resistivity and slowness model cross-gradient sensitivity matrices, respectively. The cross-gradient sensitivity is the partial derivative of t with respect to the model parameters. The term β is the model weighting that compensates for the large difference (up to three orders of magnitude) between the slowness and logarithm of resistivity models. It should be noted that β is not in the objective function in equation 5, but it is included in equation 7 to moderate the resistivity updates because they vary

over a much smaller logarithmic range than the slowness values. For the $(k+1)$ th iteration, the model parameter becomes

$$\begin{pmatrix} \mathbf{m}_e^{k+1} \\ \mathbf{m}_s^{k+1} \end{pmatrix} = \begin{pmatrix} \mathbf{m}_e^k \exp(\Delta \mathbf{m}_e^{k+1}) \\ \mathbf{m}_s^k + \Delta \mathbf{m}_s^{k+1} \end{pmatrix}, \quad k = 1, 2, 3, \dots, N. \quad (8)$$

The model update $\begin{pmatrix} \Delta \mathbf{m}_e^{k+1} \\ \Delta \mathbf{m}_s^{k+1} \end{pmatrix}$ is obtained by the CG method. CG is the preferred inversion method because quite a large number of model parameters is involved in 2D tomography imaging (Rawlinson and Sambridge, 2003) and joint inversion. Because the storage of the sensitivity matrix (of equation 7) can be avoided, the CG algorithm is computationally affordable (Sadiku, 2009).

Joint inversion workflow

Figure 1 shows the joint inversion of the seismic traveltimes and frequency-domain airborne EM data workflow implemented in this study. It is divided into two phases: the preparatory phase and the joint inversion loop phase. In the preparatory phase, the files containing the traveltimes and EM data and the initial velocity and resistivity models, and all the aforementioned weights (data, data misfit, model, roughening, and cross gradient) are specified. The two criteria that terminate the iterative inversion process, i.e., the maximum number of iterations and the threshold value for the objective function (equation 5), are also defined in this phase. The first stage in the joint inversion iteration loop is the computation of the forward responses $\mathbf{G}_e(\mathbf{m}_e)$ and $\mathbf{G}_s(\mathbf{m}_s)$ of the initial models $\mathbf{m}_e$ and $\mathbf{m}_s$. The matrix $\mathbf{G}_e(\mathbf{m}_e)$ is computed with equation 1, and $\mathbf{G}_s(\mathbf{m}_s)$ is the forward response from Zhang and Toksöz (1998). The pseudo-2D resistivity model $\mathbf{m}_e$ is interpolated to match the 2D velocity model $\mathbf{m}_s$ to have the same number of model parameters for the computation of the cross-gradient function. The sensitivity and misfits are calculated from these responses. Next, the cross-gradient t and its sensitivities $\mathbf{B}_e$ and $\mathbf{B}_s$ are calculated between $\mathbf{m}_s$ and the interpolated $\log(\mathbf{m}_e)$. To encourage the convergence of the misfit functional to the global minimum (Zhdanov, 2015), adaptive rescaling of the cross-gradients weighting λ is done via this procedure: compressing the right side of equation 7 to

$$rhs_J = \begin{pmatrix} \beta \mathbf{q}_e - \lambda \mathbf{B}_e^T t \\ \mathbf{q}_s - \lambda \mathbf{B}_s^T t \end{pmatrix}, \quad (9a)$$

where $\mathbf{q}_e = ((\mathbf{W}_e \mathbf{A}_e)^T \mathbf{W}_e \Delta \mathbf{d}_e - \tau_e \mathbf{L}^T \mathbf{Lm}_e^k)$ and $\mathbf{q}_s = ((\mathbf{W}_s \mathbf{A}_s)^T \mathbf{W}_s \Delta \mathbf{d}_s - \tau_s \mathbf{L}^T \mathbf{Lm}_s^k)$.

The cross-gradient scaling factor λ_f is defined by

$$\lambda_f = \frac{\min(\|\mathbf{q}_e\|_2, \|\mathbf{q}_s\|_2)}{\|\mathbf{B}_e^T t + \mathbf{B}_s^T t\|_2}. \quad (9b)$$

During the joint inversion, the cross-gradient weight λ in equation 7 is automatically multiplied by the scaling factor λ_f in equation 9b to normalize the different units mixed together in the scheme.

The corrections to the initial models $\begin{pmatrix} \Delta \mathbf{m}_e^{k+1} \\ \Delta \mathbf{m}_s^{k+1} \end{pmatrix}$ are then calculated by solving equation 7 by the CG method. New model parameters are calculated using equation 8. After each iteration, the value of the computed objective function is compared with the threshold

value: The iterative inversion is halted if the calculated value is lower than the threshold value ψ ; otherwise, it continues up to the predefined maximum number of iterations. We now demonstrate how the methodology is used first on synthetic, then on field data.

SYNTHETIC EXPERIMENTS

We demonstrate the joint inversion concept described in the previous sections on synthetic data computed from velocity and resistivity models with similar structures (Figure 2). Following an

approach similar to Gallardo and Meju (2004), the velocity model consists of two anomalous blocks (60 m deep) embedded in the velocity-gradient background (the velocity gradient background corresponds to the layered earth assumption made previously): The low-velocity block on the right side is 2600 m/s, whereas the high-velocity block on the left is 3000 m/s. The resistivity model has a similar structure to the velocity model with 177.8 and 309.7 Ωm resistivity values for the right and left blocks, respectively. This is the kind of model expected if velocity and resistivity are controlled by, say, realistic water saturations and rock porosity or ma-

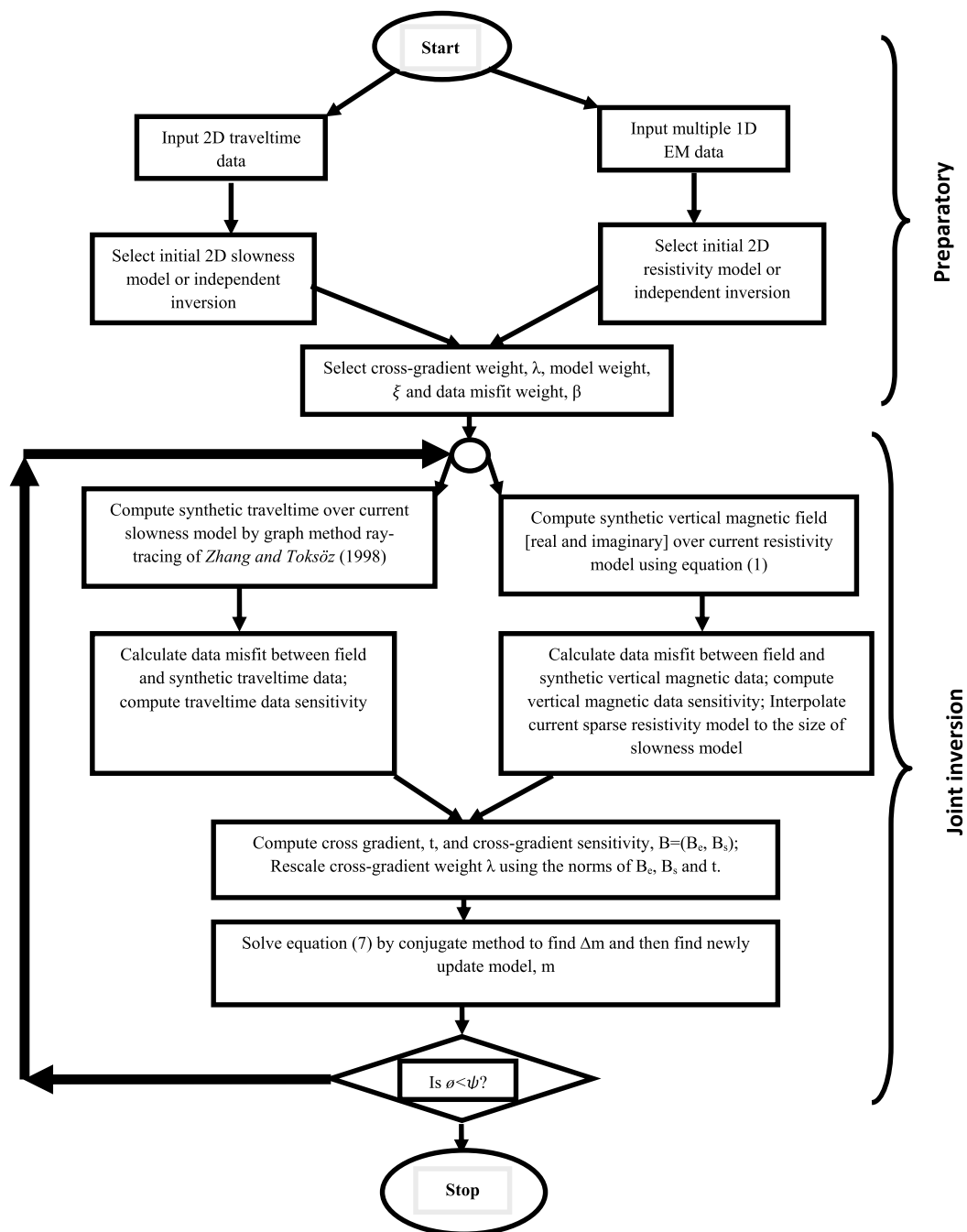


Figure 1. Joint inversion workflow, consisting two phases: the preparatory phase and the joint inversion loop. The joint inversion iteration stops either at the maximum iteration set or when the objective function ϕ is less than a threshold value ψ .

trix. We compute the first-arrival times for 360 shots and 200 receivers located at the surface of the velocity model. Shot-receiver offsets range from 0.5 to 1042.5 m. The EM responses are computed at 60 locations every 20 m along the line. The instrument altitude was fixed at 50 m above ground; the transmitter and receiver are HCP coils separated by 7.93 m, and the system generated four frequencies logarithmically spaced from 100 Hz to 100 kHz.

Except otherwise stated the traveltime and the EM data are contaminated by 1% random noise. We perform separate (independent) inversions for the traveltime and multifrequency airborne EM data. For the initial resistivity model, a 100 Ωm homogeneous value is used. The initial velocity model used is the background velocity model used to generate the true velocity model in Figure 2a. We set $\tau_s = 1$ and $\tau_e = 0.001$ to perform the independent inversions of seismic and EM data, respectively. We run the inversions for five iterations to produce the independent inverted velocity and resistivity models shown in Figure 3a and 3b, respectively. The independent velocity model inversion is unable to retrieve the relatively low-velocity anomalous block, and there is much smearing around the high-velocity block (see Figure 3a). However, from the independent frequency-domain airborne EM data inversion, the conductive and resistive blocks are retrieved (Figure 3b). Whereas the target data misfit is taken as the added noise in percentage, the recovered

data misfit (Δ_d) is defined as $\Delta_d = 100\% \sqrt{\sum \left(\frac{X_{\text{mod}} - X_{\text{obs}}}{X_{\text{obs}}} \right)^2}$, where X_{obs} is the observed data (equivalent to $\mathbf{d}_e$ or $\mathbf{d}_s$ in equation 5) and X_{mod} is the computed data from the inverted model corresponding to $G(\mathbf{m}_e)$ or $G(\mathbf{m}_s)$ in equation 5. For the target mis-

fit of 1% and for the recovered data misfits, traveltime and EM data are 1% and 3.6%, respectively.

Parameter selections for joint inversion

For easy referencing to joint inversion parameters, we recast the objective function in equation 5 in compact notations:

$$\phi(\mathbf{m}_e, \mathbf{m}_s) = \phi_e + \phi_s + \lambda \|\mathbf{t}\|^2, \quad (10)$$

where $\phi_e = (\|\mathbf{W}_e(\mathbf{d}_e - \mathbf{G}_e(\mathbf{m}_e))\|^2 + \tau_e \|\mathbf{Lm}_e\|^2)$ and $\phi_s = (\|\mathbf{W}_s(\mathbf{d}_s - \mathbf{G}_s(\mathbf{m}_s))\|^2 + \tau_s \|\mathbf{Lm}_s\|^2)$.

Because τ_e , τ_s , λ , β , $\mathbf{W}_e$, and $\mathbf{W}_s$ are the factors that influence the inversion results (see equation 7), they should be selected in such a way that they do not overshadow or even suppress the effect of the cross-gradient constraint weight λ during the joint inversion process. In this section, we explain the different weights in equation 7 and how we choose them for optimum performance in our synthetic example. In the previous section, the model regularization factors (τ_s , τ_e) chosen help the nonlinear inversion to converge by stabilizing the process. The model weighting factor β acts as a magnitude scaling factor for $\log(\mathbf{m}_e)$ and $(\mathbf{m}_s)$. The expression used for computing the model weighting factor is $\beta = \log(\max(\mathbf{m}_e)) * \min(\mathbf{m}_s)$. From Figure 2, $\max(\mathbf{m}_e) = 309.9 \Omega\text{m}$ and $\min(\mathbf{m}_s) = 3.0e - 4 \text{ s/m}$. Therefore, for our synthetic example, $\beta = 0.0019$.

To select an optimum cross-gradient weight (λ) for the joint inversion, we perform joint inversions with several values of λ . The

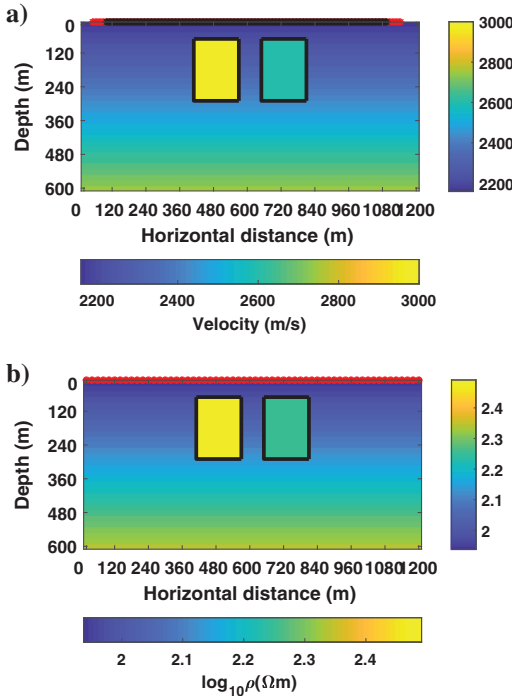


Figure 2. (a) True velocity model with 220 sources and 200 receivers on the surface, and (b) true resistivity model, with 60 data points on the surface of the model. The red and black dots in (a) are, respectively, source and receiver positions; the red dots in (b) are the EM sounding locations.

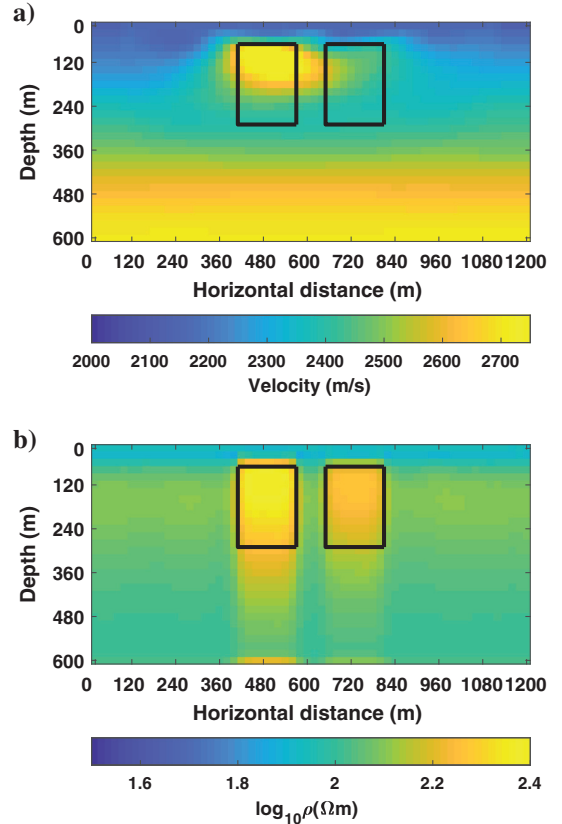


Figure 3. Independent inversions for traveltime and frequency-domain airborne EM data after five iterations: inverted (a) velocity and (b) resistivity models.

values tested for λ are every half-power of 10 between 0.1 and 10,000 units. Using the independent results in Figure 3 as initial models, we perform joint inversions for additional maximum of

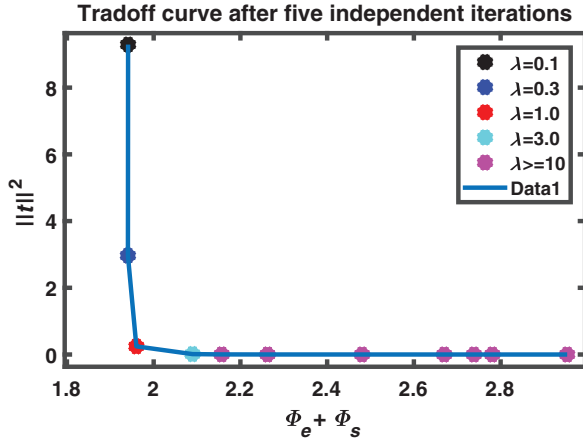


Figure 4. Tradeoff curve from the plot of $\phi_e + \phi_s$ against the cross-gradient (t) norm for 11 cross-gradient weights (λ) from 0.1 to 10,000 units. The joint inversion is preceded by five iterations of independent inversions. The values ϕ_e and ϕ_s are defined in the text.

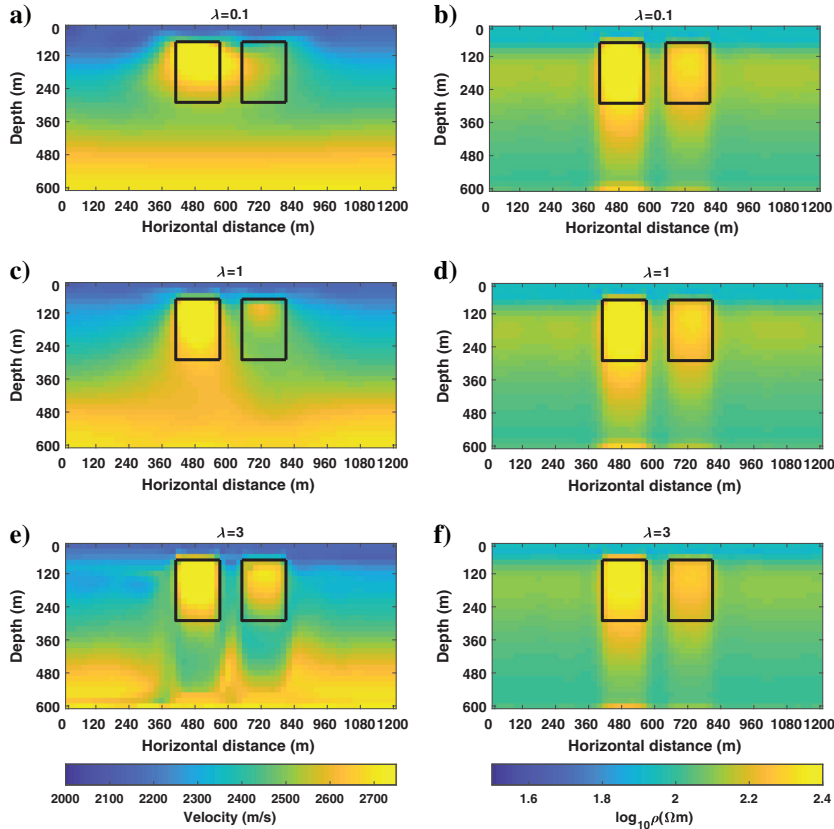


Figure 5. Jointly inverted velocity (left column) and resistivity (right column) models for different cross-gradient weights (λ). The joint inversion is preceded by five iterations of independent inversions. (a and b) The lowest cross-gradient weight is $\lambda = 0.1$; (c and d) it is increased to $\lambda = 1$ in. The high cross-gradient weights (e and f) dominate data and model misfits increasing structural similarity with the resistivity models.

10 iterations or until there is no appreciable change between consecutive norms of the cross gradient t . Similar to the usual tradeoff curve from model and data misfit norms, we plot $\phi_e + \phi_s$ against t^2 , in equation 10, for the 11 values of λ . Figure 4 shows the tradeoff curve produced, which suggests that the sum $\phi_e + \phi_s$ increases directly with λ . The optimum cross gradient from the tradeoff curve is $\lambda = 3$. However, for a stronger structural similarity, a value of $\lambda > 3$ can be used. Of course, using a large λ may result in larger data and model misfit norms.

In Figure 5, we show the velocity and resistivity models from the joint inversion for $\lambda = 0.1, 1$, and 3. Comparing Figure 3a with the velocity model in Figure 5 for $\lambda = 0.1$, we note that some part of the left resistivity structure has been communicated to the left velocity block, although the smearing still exists. But the magnitude of $\lambda = 0.1$ is not strong enough to indicate structural similarity between the right velocity and the right resistive blocks. For a target data misfit of 1% (noise level), the recovered traveltime and EM data misfits are, respectively, 1% and 3.6%. For $\lambda = 1$, the left velocity block becomes more focused at the edge as some velocity structure begins to appear in the right block (with recovered traveltime and EM data misfits of 1% and 2%, respectively, over a target data misfit of 1%). When λ becomes 3, the structural similarities between the two models increase with greater focusing resolution (with recovered traveltime and EM data misfits of 1.4% and 3.2%, respectively, over a target data misfit of 1%). It is noteworthy that recovered data misfits increase as the structural similarity or cross-gradient weight (λ) increases.

As the cross gradient becomes dominant over model and data misfit norms (at higher λ), the increased structural similarities of the velocity and the resistivity models are summarized in Figure 6. Figure 6 shows data rms misfit (frequency-domain airborne EM and traveltime), normalized cross-gradient rms (cross-gradients rms normalized by the first cross-gradient rms value at iteration 6), and the objective function in equation 5 for the different λ s. The cross-gradient rms is computed with

$$t_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (t_i)^2}, \quad (11)$$

where N is the number of model parameters. It can be observed from Figure 6 that there is a tradeoff between increasing structural similarity (minimizing equation 11) and minimizing the data (frequency-domain airborne EM and traveltime) at very high λ values.

Initial model selections for joint inversion

In the “Parameter selections for joint inversion” subsection, a maximum of 10 joint inversion iterations are preceded by five independent inversion iterations. We argued that this prevents the computation of near-zero cross gradients from models that have very little structural content in the first place. This computation would not only be misleading, it would also prevent

the cross-gradient part of the objective function from updating. In this subsection, we present joint inversion starting from the initial models used to produce independent models in Figure 3. We run two independent inversions at the first iteration and then we begin the joint inversion at iteration 2. All parameters used for the previous experiment are held constant except that we start from the initial models without any structures. After 10 iterations, setting $\lambda = 3$ (optimum value from the previous experiment), the jointly inverted velocity and resistivity models are shown in Figure 7. The structural exchange begins too early when the data misfits are high (recovered data misfits are, respectively, 2% and 7.9% for the traveltime and EM data). In other words, jointly inverted models for the simultaneously optimum cross-gradient rms and data misfits are sought for. This eventually outputs velocity and resistivity models with lower resolutions than those in Figure 5. Thus, it is remarkably noted that higher resolution models can be retrieved when the initial models for the joint inversion are from independent inversions, where the data misfits have been minimized to a certain level, than using homogenous ones (comparing the smaller recovered data misfits in Figure 5 [1.4% for traveltime and 3.2% for EM data] with the high values [2% for traveltime and 7.9% for EM data] in Figure 7).

Signal-to-noise and resolution

Until now, we have kept the noise level to 1%. To learn about the effects of noise on the resolution of the jointly inverted models, we vary the noise magnitudes from 1% to 5% keeping other optimum factors constant from the previous subsections. Figure 8 shows that as the noise magnitudes increase, the resolutions of the blocks decrease. At 5% noise level, the block on the right hand appears to be fading out. The rms plots in Figure 9 confirm this with the systematic increase in data misfits with the noise levels. However, the behavior of the structural similarity stays the same for all cases because the available structures cross talked in the same way. For 1% target data misfits, the recovered data misfits are, respectively, 1.4% and 3.2% for the traveltime and EM data. The recovered data misfits over the 3% target data misfit are 3.1% and 4.2% for the traveltime and EM data, respectively. Last for the 5% target data misfit, we recovered 5.1% and 5.5% data misfits for the traveltime and EM data, respectively.

Our synthetic experiments show that the optimum results from the joint inversion should be preceded by independent traveltime and frequency-domain EM data inversion. Also, the choice of the cross-gradient weight should be made with the awareness of the tradeoff between minimizing the data misfits and enhancing the structural similarity between the velocity and resistivity models. In comparison with the results in Gallardo and Meju (2004), where the cross-gradient weight was reduced as iteration progressed. Expectedly, the data misfits were minimized as the structural similarity was relaxed with iteration. Last, although in the presence of random noise, the cross-structural communication is not affected, a

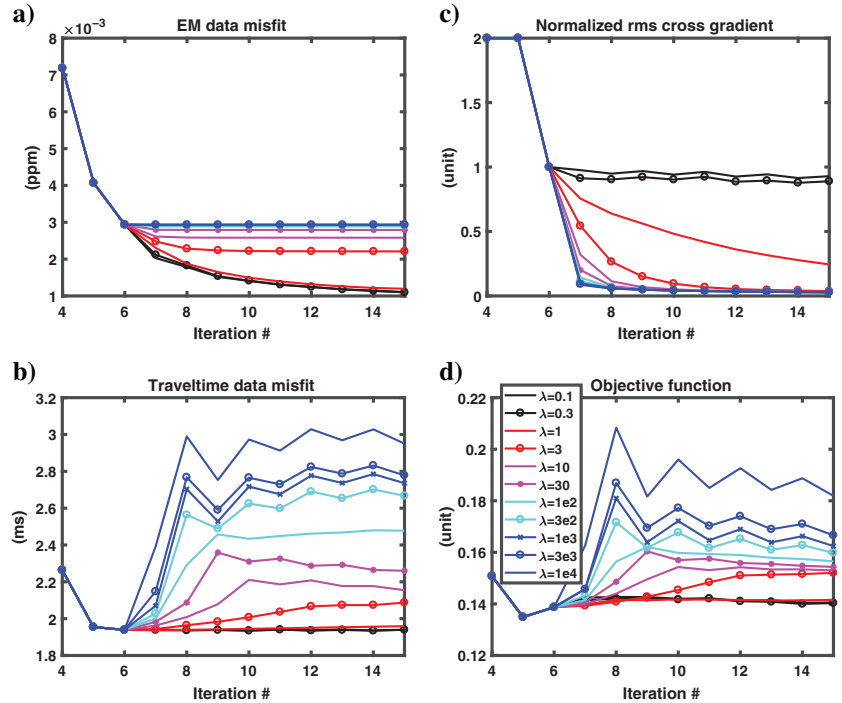


Figure 6. (a) Frequency-domain airborne EM data rms misfit, (b) traveltime rms misfit, (c) normalized cross-gradient rms, and (d) objective function at various cross-gradient weights (λ). The first cross-gradient starts at iteration 6 (after five iterations for the frequency-domain airborne EM and traveltime independent inversions), but all models vary the same way, hence the overlapping of curves.

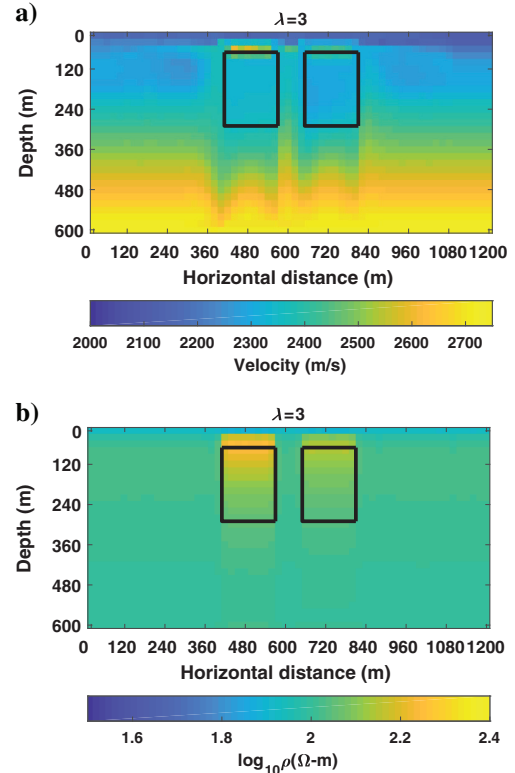


Figure 7. Jointly inverted (a) velocity and (b) resistivity models for the cross-gradient weight of $\lambda = 3$. The joint inversion is started from homogeneous initial models.

poor signal-to-noise ratio could diminish the resolution of the jointly inverted models.

JOINT INVERSION OF FIELD DATA

The field data are from coincident seismic and frequency-domain airborne EM surveys in Canada. The near surface of the geology in the survey area consists essentially of recent Quaternary post-glacial deposits, sands, tills, and moraines.

We jointly invert collocated velocity and resistivity models across an 8 km-long survey profile. The seismic survey consists of 19 dynamite shots and 140 receivers. Shot spacing is about 400 m and receiver spacing is 50 m. The high quality of the data allows an easy first-arrival picking.

The frequency-domain airborne EM data are the HCP in-phase and quadrature responses acquired at five frequencies (390, 1787, 8391, 40,740, and 1,31,630 Hz) with Fugro (now Compagnie Générale de Géophysique) Airborne's RESOLVE system. We limited the very dense frequency-domain airborne EM data to 72 points to have a station spacing (~ 100 m) similar to the footprint of a 1 kHz VMD system flown at an altitude of 50 m over a 10 Ωm half space (Beamish, 2003).

The initial velocity model is built from the first breaks. An independent traveltime tomography inversion is performed to minimize the traveltime misfits. The result of the independent velocity

inversion is shown in Figure 10a. We then run an independent inversion of the airborne EM data using the pseudo-2D method presented in subsection "Pseudo-2D frequency-domain airborne EM inversion" using a 30 Ωm half-space as the initial model. The resistivity model resulting from the independent inversion is shown in Figure 10b. The models in Figure 10 are then used as the initial models for the joint inversion workflow of Figure 2 setting the cross-gradient weight λ to 10.

But let us first return to the inversion process. The evolution of the EM and traveltime data misfits, as well as that of the normalized (to half of its value at the first iteration) cross-gradient rms, is shown in Figure 11. Note that we run one more iteration of independent inversions for each data set before turning on the cross-gradient constraint at iteration 2. The frequency-domain airborne EM rms misfit drops rapidly when the cross-gradient constrain is turned on, and then it remains almost constant throughout the inversion (Figure 11a). The traveltime rms misfit, however, decreases more smoothly, with a drop of only 0.4 ms after 30 iterations (Figure 11b). The cross-gradient rms follows a less stable pattern and even increases at a few iterations. It nonetheless decreases significantly by a factor of approximately 4 between iterations 2 (i.e., when it is turned on) and 30, indicating that indeed structural similarity increases as the joint inversion progresses.

The velocity models shown at Figure 12a (independent tomog-

raphy) and Figure 12b (joint inversion) appear to be fairly similar, but these have several differences, essentially in the top 100 m where the airborne EM data have the most sensitivity. The velocity at the top 50 m clearly decreases, and we observe velocity increases right below this top low-velocity layer. The resistivity model (Figure 12c) also undergoes noticeable changes where the resistive and conductive blocks produced by the independent EM inversion (Figure 10) have been pushed deeper. These small but noticeable changes confirm the relevance of the joint inversion approach: The frequency-domain airborne EM data have more influence on the shallow part of the model, and the seismic data have more influence on its deeper part.

The main objective of the joint inversion is to produce a more reliable near-surface velocity model that can be used to compute more reliable static corrections. Velocity models from traveltime tomography inversion (Figure 10a) and from joint inversion (Figure 12b) are used to calculate statics corrections. A comparison of the traveltime-only and joint inversion statics is shown in Figure 13. The statics corrections show some noticeable differences, up to 10 ms.

The data were processed to produce stack sections following a standard processing sequence:

- geometry mapping along traverse line
- t^2 scaling
- tapers at start (20 ms) and end (500 ms) of each trace
- predictive deconvolution (16 ms lag, 160 ms operator length, 0.1% pre-whitening)

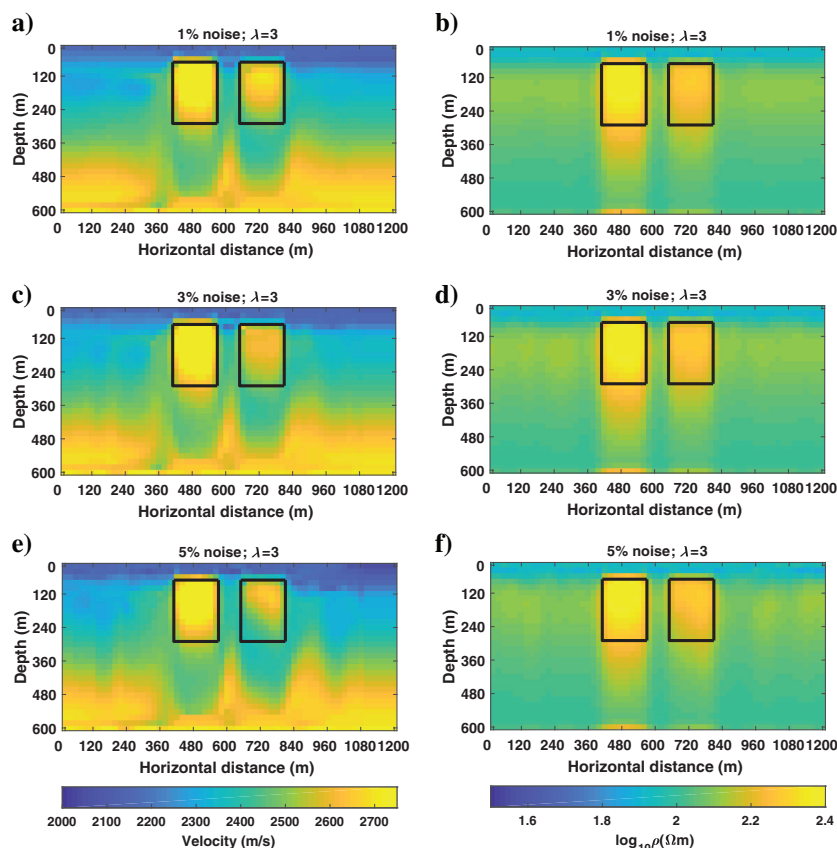


Figure 8. The effect of noise on the resolution of the inverted velocity (left column) and resistivity (right column) models for the optimum cross-gradient weights $\lambda = 3$. The low S/N reduces the resolution of the inverted velocity block, as noticed in Figure 8e for a noise magnitude of 5%.

- spectral whitening 8–75 Hz
- statics computed from inversion velocity model
- dip filter
- common mid-point gathering
- semblance velocity analysis
- normal moveout (NMO) correction
- stack.

Excerpts from the stack sections produced with these statics corrections are presented in Figure 14a (traveltime-only) and 14b (joint inversion). The joint inversion brings some noticeable improvements in the stack:

- Small high-frequency reflectors near 1.2 s are much better resolved with joint inversion statics.
- A series of reflectors running across the section between approximately 1.3 and 1.5 s traveltime is also better focused with the joint inversion statics.
- The deeper reflectors between 1.8 and 2.1 s show a better lateral continuity with the joint inversion statics, although the improvement there is not as remarkable as for the shallower reflectors.

DISCUSSION AND PERSPECTIVES

Our first result of the application of joint inversion of seismic and airborne electromagnetic to a field data set illustrates the potential of the approach. We point out that the seismic data were acquired with a true 3D geometry, and therefore we did not have an actual 2D data set for which the improvements might have been more remarkable. So, one of the obvious perspectives for further research is to extend the methodology to 3D. As stated earlier, the approach we presented here for 2D EM modeling can easily be extended to 3D and, furthermore, the computation effort required scales almost linearly with the number of airborne EM data points. This would definitely not be the case if an actual 3D code (say, using finite elements, e.g., Raiche et al., 2007) was used.

We want to point out also that in this work, we have not made use of petrophysical relations between seismic velocity and electrical resistivity — an approach that has been shown to be successful in some cases (Colombo and Keho, 2010). Although the use of this a priori constraint could have helped the joint inversion, we show here that starting the joint inversion with models that already honor the data, though not in an optimal sense, gives satisfactory results, at least for the synthetic and real cases studied here.

The choice of smoothness parameters, as well as the data, model, and cross-gradient weights, (equation 5) gives a great number of degrees of freedom on how to design the objective function. The parameters chosen for the examples shown above are from trial-and-error while trying to ensure that the cross gradient has sufficient weight to force the velocity and resistivity models to exchange structural information. There is arguably some degree of tradeoff

between some of these five parameters. We know that the model smoothness (τ) will definitely have an effect on the data fit, see Constable et al. (1987) seminal paper, but obviously also on the model's gradient. This will in turn affect the cross gradient that will need to be weighted accordingly.

One interesting observation in Figure 11 is that although the EM misfit remains stable from iteration 5, the traveltime misfit keeps on decreasing, although by small amounts. Nonetheless, this evolution pattern suggests that it is the resistivity model that drives the joint

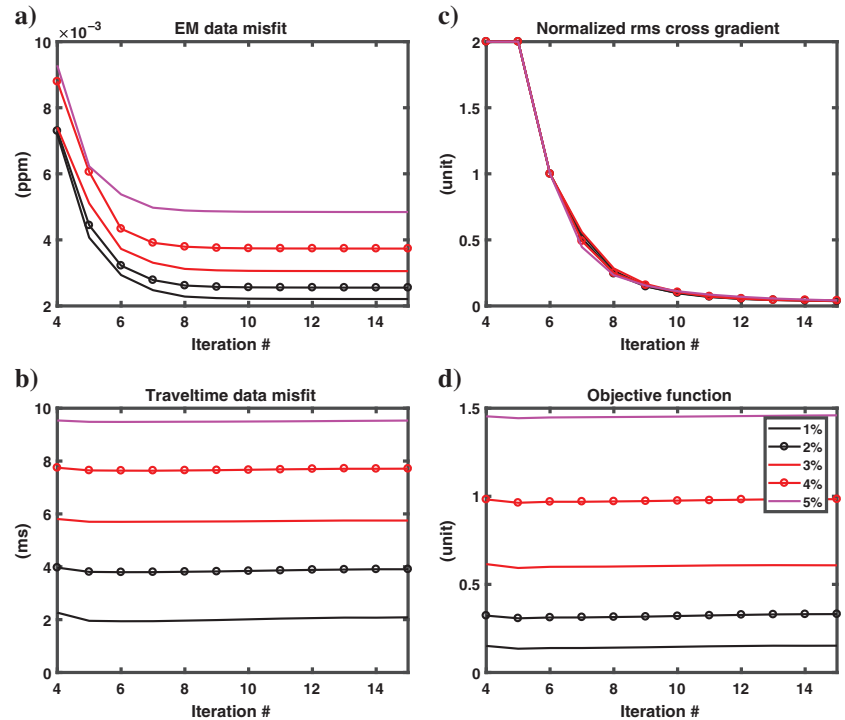


Figure 9. (a) Frequency-domain airborne EM data rms misfit, (b) traveltime rms misfit, (c) normalized cross-gradient rms, and (d) objective function for cross-gradient weights $\lambda = 3$ as a function of the increasing noise magnitude: 1% to 5%. Although the behavior of the structural similarity exchange remains similar at all noise levels (c), the misfits (a, b, and d) increase as functions of the noise magnitudes.

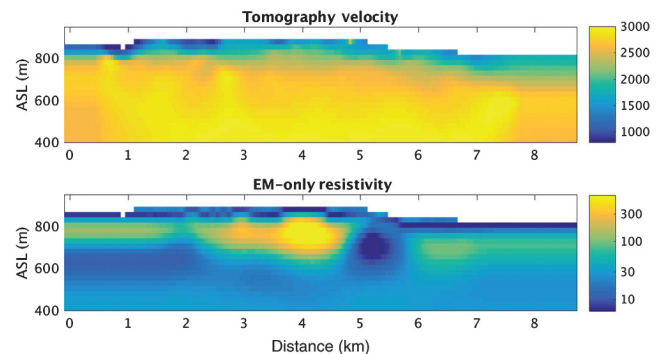


Figure 10. (a) Inverted velocity model from independent traveltime inversion and (b) inverted resistivity model from independent resistivity inversion.

inversion; i.e., it brings the velocity model structure closer to its own. This might also be related to the relatively small dynamic of $\log(\text{resistivity})$ compared with that of the seismic velocity. A

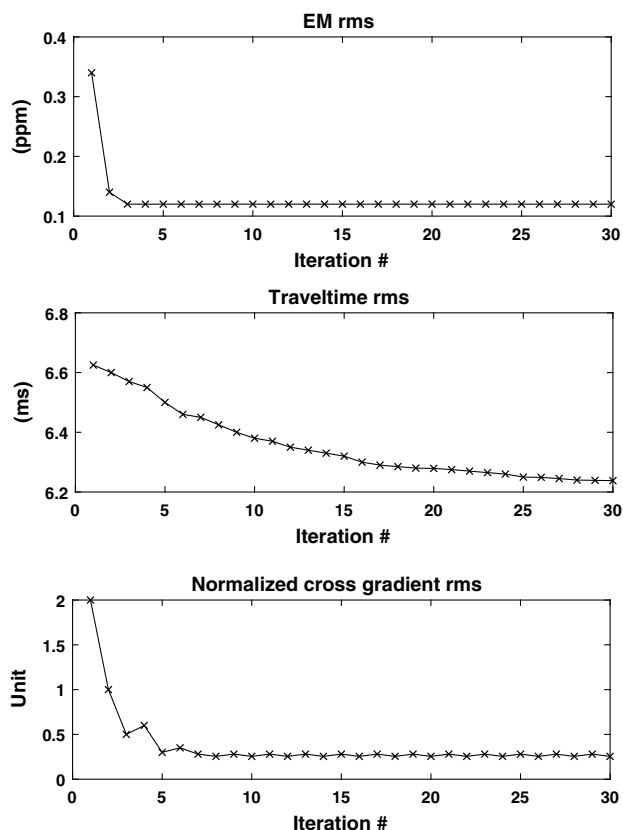


Figure 11. Global field data misfit (a) frequency-domain airborne EM, (b) traveltime, and (c) normalized cross-gradient rms. The cross-gradient constrained joint inversion begins at iteration 2.

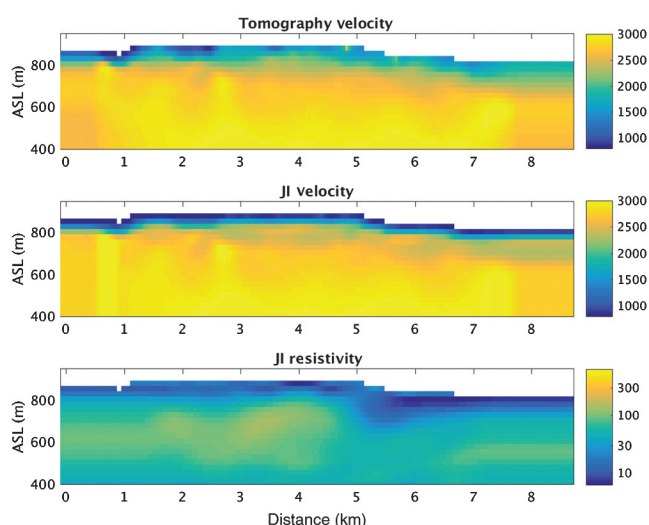


Figure 12. (a) Traveltime tomography velocity model used as a starting model for the joint inversion, (b) final velocity model from joint inversion, and (c) inverted resistivity model from joint inversion.

change of, say, 1 unit of $\log(\text{resistivity})$ has a much greater effect on the forward response than the same change in velocity. We arbitrarily adjust for this difference in sensitivity scale by adjusting the model weights (see parameter β , above) based on the parameter range but adjusting in the sensitivity domain might be a more realistic (however, tedious) approach.

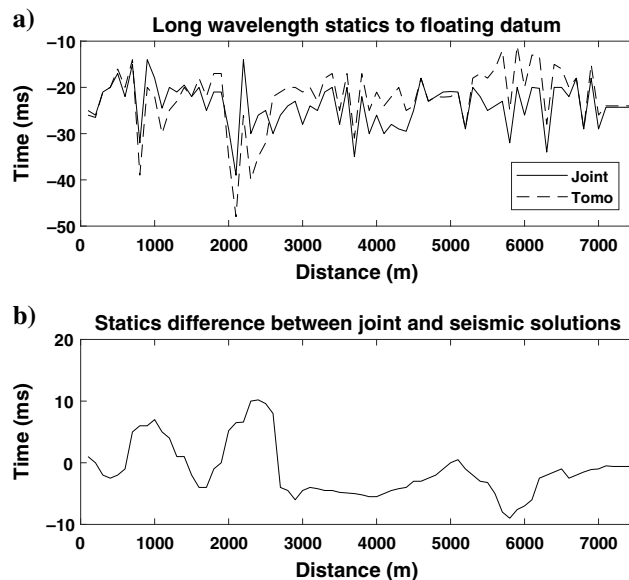


Figure 13. (a) statics comparison between independent tomography (dashed) and joint inversion (solid). (b) The difference between the two static corrections.

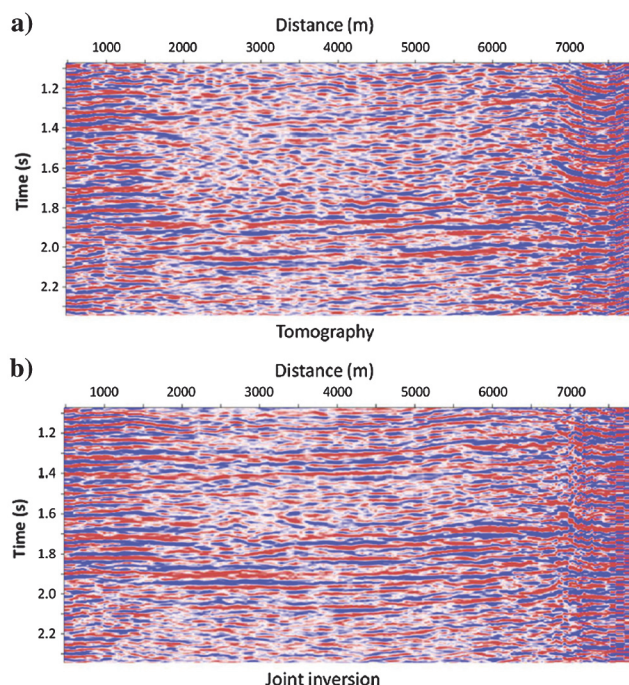


Figure 14. Portion of the stack section produced by applying statics from (a) independent traveltime inversion and (b) joint seismic-airborne EM inversion. Note the enhanced reflector continuity near 1.4 s and between 1.6 and 1.8 s.

CONCLUSION

With a two-phase workflow for structurally constrained joint inversion of seismic and EM data, we establish mathematical expressions and procedure for selecting optimum parameters for the process. From the various synthetic experimentations for different initial models, S/N ratio and cross-gradient weights we are able to establish the means of parameter selections. Because the recovered data misfits increase with the cross-gradient weights, we note that there is a tradeoff between minimizing the data misfits and enhancing the structural similarity between the velocity and resistivity models. Also, it is recommended to start the joint inversion with an initial model with some structure rather than using a homogeneous model. Although the amount of noise reflects in the data misfits, the effects gradually affect the structural semblance at a low S/N. We have used the results from the joint inversion of 2D seismic data and pseudo-2D frequency-domain airborne EM data to improve the near-surface velocity model for the computation of better statics corrections and seismic stack section.

Because we have assumed that the subsurface geology considered is horizontal (hence pseudo-2D from multiple 1D EM data), our methodology could easily be applied to data (seismic and EM) acquired in the sedimentary environment. The disadvantage is that it could not be generalized for other geologic settings. The consequent challenge will be to perform more thorough experiments for selecting optimum parameters for the joint inversion in such a geologic setting. Therefore, the future work should include extending the concept to a general geologic setting other than a sedimentary environment.

ACKNOWLEDGMENTS

We gratefully acknowledge the financial support of the National Natural Science Foundation of China (grant no. 41374132). Thanks to GeoTomo for making available their software for this project and to Shell Canada for granting us permission to show their data. We acknowledge the use of frequency-domain airborne EM forward code developed by A. Raiche for AMIRA project P223F. Comments by P. Lelièvre and two anonymous reviewers greatly helped us improve this manuscript.

REFERENCES

- Abubakar, A., G. Gao, T. M. Habashy, and J. Liu, 2012, Joint inversion approaches for geophysical electromagnetic and elastic full-waveform data: *Inverse Problems*, **28**, 055016, doi: [10.1088/0266-5611/28/5/055016](https://doi.org/10.1088/0266-5611/28/5/055016).
- Anderson, W. L., 1979, Numerical integration of related Hankel transforms of orders 0 and 1 by adaptive digital filtering: *Geophysics*, **44**, 1287–1305, doi: [10.1190/1.1441007](https://doi.org/10.1190/1.1441007).
- Auken, E., and A. V. Christiansen, 2004, Layered and laterally constrained 2D inversion of resistivity data: *Geophysics*, **69**, 752–761, doi: [10.1190/1.1759461](https://doi.org/10.1190/1.1759461).
- Auken, E., F. Jørgensen, and K. I. Sørensen, 2003, Large-scale TEM investigation for groundwater: *Exploration Geophysics*, **34**, 188–194, doi: [10.1071/EG03188](https://doi.org/10.1071/EG03188).
- Beamish, D., 2003, Airborne EM footprints: *Geophysical Prospecting*, **51**, 49–60, doi: [10.1046/j.1365-2478.2003.00353.x](https://doi.org/10.1046/j.1365-2478.2003.00353.x).
- Brown, V., K. Key, and S. Singh, 2012, Seismically regularized controlled-source electromagnetic inversion: *Geophysics*, **77**, no.1, E57–E65, doi: [10.1190/geo2011-0081.1](https://doi.org/10.1190/geo2011-0081.1).
- Carter-McAuslan, A., P. G. Lelièvre, and C. G. Farquharson, 2015, A study of fuzzy c-means coupling for joint inversion, using seismic tomography and gravity data test scenarios: *Geophysics*, **80**, no. 1, W1–W15, doi: [10.1190/geo2014-0056.1](https://doi.org/10.1190/geo2014-0056.1).
- Chen, J., and G. M. Hoversten, 2012, Joint inversion of marine seismic AVA and CSEM data using statistical rock-physics models and Markov random fields: *Geophysics*, **77**, no.1, R65–R80, doi: [10.1190/geo2011-0219.1](https://doi.org/10.1190/geo2011-0219.1).
- Colombo, D., and T. Keho, 2010, The non-seismic data and joint inversion strategy for the near surface solution in Saudi Arabia: 80th Annual International Meeting, SEG, Expanded Abstracts, 1934–1938, doi: [10.1190/1.3513222](https://doi.org/10.1190/1.3513222).
- Colombo, D., G. McNeice, and R. Ley, 2012, Multigeophysics joint inversion for complex land seismic imaging in Saudi Arabia: 82nd Annual International Meeting, SEG, Expanded Abstracts, doi: [10.1190/segam2012-0441.1](https://doi.org/10.1190/segam2012-0441.1).
- Colombo, D., G. McNeice, D. Rovetta, E. Turkoglu, A. Sena, E. Sandoval-Curiel, F. Miorrelli, and Y. Taqi, 2015, Super resolution multi-geophysics imaging of a complex wadi for near surface corrections: 85th Annual International Meeting, SEG, Expanded Abstracts, 849–853, doi: [10.1190/segam2015-5841055.1](https://doi.org/10.1190/segam2015-5841055.1).
- Colombo, D., G. McNeice, D. Rovetta, E. Sandoval-Curiel, E. Turkoglu, and A. Sena, 2016, High-resolution velocity modelling by seismic-airborne TEM joint inversion: A new perspective for near-surface characterization: *The Leading Edge*, **35**, 977–985, doi: [10.1190/le35110977.1](https://doi.org/10.1190/le35110977.1).
- Constable, S., and L. J. Srnka, 2007, An introduction to marine controlled-source electromagnetic methods for hydrocarbon exploration: *Geophysics*, **72**, no. 2, WA3–WA12, doi: [10.1190/1.2432483](https://doi.org/10.1190/1.2432483).
- Constable, S. C., R. L. Parker, and C. G. Constable, 1987, Occam's inversion: A practical algorithm for generating smooth models from electromagnetic data: *Geophysics*, **52**, 289–300, doi: [10.1190/1.1442303](https://doi.org/10.1190/1.1442303).
- Eidesmo, T., S. Ellingsrud, L. M. MacGregor, S. Constable, M. C. Sinha, S. E. Johansen, F. N. Kong, and H. Westerdahl, 2002, Sea bed logging (SBL), a new method for remote and direct identification of hydrocarbon filled layers in deepwater areas: *First Break*, **20**, 144–152.
- Ernst, F. E., 2007, Long-wavelength statics estimation from guided waves: 69th Annual International Conference and Exhibition, EAGE, Extended Abstracts, E033, doi: [10.3997/2214-4609.201401612](https://doi.org/10.3997/2214-4609.201401612).
- Farquharson, C. G., and D. W. Oldenburg, 1996, Approximate sensitivities for the electromagnetic inverse problem: *Geophysical Journal International*, **126**, 235–252, doi: [10.1111/j.1365-246X.1996.tb05282.x](https://doi.org/10.1111/j.1365-246X.1996.tb05282.x).
- Farquharson, C. G., D. W. Oldenburg, and P. S. Routh, 2003, Simultaneous 1D inversion of loop-loop electromagnetic data for magnetic susceptibility and electrical conductivity: *Geophysics*, **68**, 1857–1869, doi: [10.1190/1.1635038](https://doi.org/10.1190/1.1635038).
- Gallardo, L. A., 2004, Joint two-dimensional inversion of geoelectromagnetic and seismic refraction data with cross-gradients constraint: Ph.D. thesis, Lancaster University.
- Gallardo, L. A., and M. A. Meju, 2003, Characterization of heterogeneous near-surface materials by joint 2D inversion of DC resistivity and seismic data: *Geophysical Research Letters*, **30**, 1658, doi: [10.1029/2003GL017370](https://doi.org/10.1029/2003GL017370).
- Gallardo, L. A., and M. A. Meju, 2004, Joint two-dimensional DC resistivity and seismic travel time inversion with cross-gradients constraints: *Journal of Geophysical Research*, **109**, B03311, doi: [10.1029/2003JB002716](https://doi.org/10.1029/2003JB002716).
- Gao, G., A. Abubakar, and T. M. Habashy, 2010, Simultaneous joint petrophysical inversion of electromagnetic and seismic measurements: 80th Annual International Meeting, SEG, Expanded Abstracts, 2799–2804, doi: [10.1190/1.3513425](https://doi.org/10.1190/1.3513425).
- Gao, G., A. Abubakar, and T. M. Habashy, 2012, Joint petrophysical inversion of electromagnetic and full-waveform seismic data: *Geophysics*, **77**, no. 3, WA3–WA18, doi: [10.1190/geo2011-0157.1](https://doi.org/10.1190/geo2011-0157.1).
- Gardner, L. W., 1967, Refraction seismograph profile interpretation, in A. W. Musgrave, ed., *Seismic refraction prospecting*: SEG, 338–347.
- Garofalo, F., G. Sauvin, L. V. Socco, and I. Lecomte, 2015, Joint inversion of seismic and electric data applied to 2D media: *Geophysics*, **80**, no. 4, EN93–EN104, doi: [10.1190/geo2014-0313.1](https://doi.org/10.1190/geo2014-0313.1).
- Hampson, D., and B. Russell, 1984, First-break interpretation using generalized linear inversion: *Canadian Journal of Exploration Geophysics*, **20**, 40–54.
- Hatherly, P., M. Urosevic, A. Lambourne, and B. J. Evans, 1994, A simple approach to calculating refraction statics corrections: *Geophysics*, **59**, 156–160, doi: [10.1190/1.1443527](https://doi.org/10.1190/1.1443527).
- Hesthammer, J., A. Stefatos, and S. Sperrevik, 2012, CSEM efficiency — Evaluation of recent drilling results: *First Break*, **30**, 47–55, doi: [10.3997/1365-2397.2012007](https://doi.org/10.3997/1365-2397.2012007).
- Hill, R., 1987, Downward continuation of refracted arrivals to determine shallow structures: *Geophysics*, **52**, 1188–1198, doi: [10.1190/1.1442382](https://doi.org/10.1190/1.1442382).
- Hole, J. A., and B. C. Zelt, 1995, 3-D finite difference reflection traveltimes: *Geophysical Journal International*, **121**, 427–434, doi: [10.1111/j.1365-246X.1995.tb05723.x](https://doi.org/10.1111/j.1365-246X.1995.tb05723.x).
- Hoversten, G. M., F. Cassassuce, E. Gasperikova, G. A. Newman, J. Chen, Y. Rubin, Z. Hou, and D. Vasco, 2006, Direct reservoir parameter estimation using joint inversion of marine seismic AVA and CSEM data: *Geophysics*, **71**, no. 3, C1–C13, doi: [10.1190/1.2194510](https://doi.org/10.1190/1.2194510).
- Hu, W., A. Abubakar, and T. M. Habashy, 2009, Joint electromagnetic and seismic inversion using structural constraints: *Geophysics*, **74**, no. 6, R99–R109, doi: [10.1190/1.3246586](https://doi.org/10.1190/1.3246586).

- Jegen, M. D., R. W. Hobbs, P. Tarits, and A. Chave, 2009, Joint inversion of marine magnetotelluric and gravity data incorporating seismic constraints: Preliminary results of sub-basalt imaging off the Faroe Shelf: *Earth and Planetary Science Letters*, **282**, 47–55, doi: [10.1016/j.epsl.2009.02.018](https://doi.org/10.1016/j.epsl.2009.02.018).
- Klimes, L., and M. Kvasnicka, 1993, 3-D network ray tracing: *Geophysical Journal International*, **116**, 726–738, doi: [10.1111/j.1365-246X.1994.tb03293.x](https://doi.org/10.1111/j.1365-246X.1994.tb03293.x).
- Lelièvre, P. G., C. G. Farquharson, and C. A. Hurich, 2011a, Computing first-arrival seismic traveltimes on unstructured 3-D tetrahedral grids using the fast marching method: *Geophysical Journal International*, **184**, 885–896, doi: [10.1111/j.1365-246X.2010.04880.x](https://doi.org/10.1111/j.1365-246X.2010.04880.x).
- Lelièvre, P. G., C. G. Farquharson, and C. A. Hurich, 2011b, Inversion of first-arrival seismic traveltimes without rays, implemented on unstructured grids: *Geophysical Journal International*, **185**, 749–763, doi: [10.1111/j.1365-246X.2011.04964.x](https://doi.org/10.1111/j.1365-246X.2011.04964.x).
- Lelièvre, P. G., C. G. Farquharson, and C. A. Hurich, 2012, Joint inversion of seismic traveltimes and gravity data on unstructured grids with application to mineral exploration: *Geophysics*, **77**, no. 1, K1–K15, doi: [10.1190/geo2011-0154.1](https://doi.org/10.1190/geo2011-0154.1).
- Lien, M., and T. Manneth, 2011, Structural joint inversion of AVO and CSEM data using flexible representations: 81st Annual International Meeting, SEG, Expanded Abstracts, 1897–1902, doi: [10.1190/1.3627577](https://doi.org/10.1190/1.3627577).
- Meju, M. A., 1994, *Geophysical data analysis: Understanding inverse problem theory and practice*, 1st ed.: SEG Course Notes Series 6.
- Megbel, N. M., G. D. Egbert, P. E. Wannamaker, A. Kelberta, and A. Schultz, 2014, Deep electrical resistivity structure of the northwestern U.S. derived from 3-D inversion of USArray magnetotelluric data: *Earth and Planetary Science Letters*, **402**, 290–304, doi: [10.1016/j.epsl.2013.12.026](https://doi.org/10.1016/j.epsl.2013.12.026).
- Minsley, B. J., J. D. Abraham, B. D. Smith, J. C. Cannia, C. I. Voss, M. T. Jorgenson, M. A. Walvoord, B. K. Wylie, L. Anderson, L. B. Ball, M. Deszcz-Pan, T. P. Wellman, and T. A. Ager, 2012, Airborne electromagnetic imaging of discontinuous permafrost: *Geophysical Research Letters*, **39**, L02503, doi: [10.1029/2011GL050079](https://doi.org/10.1029/2011GL050079).
- Moorkamp, M., 2017, Integrating electromagnetic data with other geophysical observations for enhanced imaging of the earth: A tutorial and review: *Surveys in Geophysics*, **38**, 935–962, doi: [10.1007/s10712-017-9413-7](https://doi.org/10.1007/s10712-017-9413-7).
- Moorkamp, M., B. Heincke, M. Jegen, A. W. Roberts, and R. W. Hobbs, 2011, A framework for 3-D joint inversion of MT, gravity and seismic refraction data: *Geophysical Journal International*, **184**, 477–493, doi: [10.1111/j.1365-246X.2010.04856.x](https://doi.org/10.1111/j.1365-246X.2010.04856.x).
- Moorkamp, M., A. G. Jones, and S. Fishwick, 2010, Joint inversion of receiver functions, surface-wave dispersion and magnetotelluric data: *Journal of Geophysical Research*, **115**, B04318, doi: [10.1029/2009JB006369](https://doi.org/10.1029/2009JB006369).
- Moser, T. J., 1991, Shortest path calculation of seismic rays: *Geophysics*, **56**, 59–67, doi: [10.1190/1.1442958](https://doi.org/10.1190/1.1442958).
- Muñoz, G., 2014, Exploring for geothermal resources with electromagnetic methods: *Surveys in Geophysics*, **35**, 101–122, doi: [10.1007/s10712-013-9236-0](https://doi.org/10.1007/s10712-013-9236-0).
- Nabighian, M. N., and J. C. Macnae, 1991, Time domain electromagnetic prospecting methods, in M. N. Nabighian, ed., *Electromagnetic methods in applied geophysics*, Vol. 2: Applications, Part A: SEG, 427–520.
- Palacky, G. J., and G. F. West, 1991, Airborne electromagnetic methods, in M. N. Nabighian, ed., *Electromagnetic methods in applied geophysics*, v. 2 — Applications, Part B, Investigations in Geophysics 3: SEG, 811–877.
- Patra, H. P., and K. Mallick, 1980, *Geosounding principles*, 2, Methods in geochemistry and geophysics: Elsevier 14B, 419.
- Pineda, A., S. Gallo, and H. Harkas, 2015, Geophysical near-surface characterization for static corrections: Multi-physics survey in Reggane Field, Algeria: 77th Annual International Conference and Exhibition, EAGE, Expanded Abstracts, doi: [10.3997/22144609.201412990](https://doi.org/10.3997/22144609.201412990).
- Qin, F., Y. Luo, K. B. Olsen, W. Cai, and G. T. Schuster, 1992, Finite difference solution of the eikonal equation along expanding wavefronts: *Geophysics*, **57**, 478–487, doi: [10.1190/1.1443263](https://doi.org/10.1190/1.1443263).
- Raiche, A., F. Suseng, and G. Wilson, 2007, Practical 3D EM inversion — P223F software suite: ASEG Extended Abstracts, 1–5, doi: [10.1071/ASEG2007ab114](https://doi.org/10.1071/ASEG2007ab114).
- Rawlinson, N., and M. Sambridge, 2003, *Advances in geophysics, seismic traveltime tomography of the crust and lithosphere*: Academic Press 46.
- Rockwell, D. W., 1967, A general wavefront method, in A. W. Musgrave, ed., *Seismic refraction prospecting*: SEG, 363–415.
- Sadiku, M. N. O., 2009, *Numerical techniques in electromagnetic with Matlab*, 3rd ed.: CRC, 710.
- Schamper, C., F. Rejiba, and R. Guérin, 2012, 1D single-site and laterally constrained inversion of multifrequency and multicomponent ground-based electromagnetic induction data — Application to the investigation of a near-surface clayey overburden: *Geophysics*, **77**, no. 4, WB19–WB35, doi: [10.1190/geo2011-0358.1](https://doi.org/10.1190/geo2011-0358.1).
- Schultz, G., and C. Ruppel, 2005, Inversion of inductive electromagnetic data in highly conductive terrains: *Geophysics*, **70**, no. 1, G16–G28, doi: [10.1190/1.1852775](https://doi.org/10.1190/1.1852775).
- Sethian, J., and A. Popovici, 1999, 3-D traveltime computation using the fast marching method: *Geophysics*, **64**, 516–523, doi: [10.1190/1.1444558](https://doi.org/10.1190/1.1444558).
- Sheng, S., A. Leeds, M. Buddensiek, and G. T. Schuster, 2006, Early arrival waveform tomography on near-surface refraction data: *Geophysics*, **71**, no. 4, U47–U57, doi: [10.1190/1.2210969](https://doi.org/10.1190/1.2210969).
- Smith, R., 2014, Electromagnetic induction methods in mining geophysics from 2008 to 2012: *Surveys in Geophysics*, **35**, 123–156, doi: [10.1007/s10712-013-9227-1](https://doi.org/10.1007/s10712-013-9227-1).
- Spies, B. R., and F. C. Frischknecht, 1991, *Electromagnetic methods in applied geophysics*, volume 2, Application, Parts A and B, in M. N. Nabighian, ed., *Investigations in Geophysics*, 4th ed.: SEG, 285–426.
- Sun, J., and Y. Li, 2012, Joint inversion of multiple geophysical data: A petrophysical approach using guided fuzzy c-means clustering: 82nd Annual International Meeting, SEG, Expanded Abstracts, doi: [10.1190/segam2012-1388.1](https://doi.org/10.1190/segam2012-1388.1).
- Taner, M. T., and F. Koehler, 1981, Surface consistent corrections: *Geophysics*, **46**, 17–22, doi: [10.1190/1.1441133](https://doi.org/10.1190/1.1441133).
- Vallée, M. A., R. S. Smith, and P. Keating, 2011, Metalliferous mining geophysics — State of the art after a decade in the new millennium: *Geophysics*, **76**, no. 4, W31–W50, doi: [10.1190/1.3587224](https://doi.org/10.1190/1.3587224).
- van Trier, J., and W. W. Symes, 1991, Upwind finite difference calculation of traveltimes: *Geophysics*, **56**, 812–821, doi: [10.1190/1.1443099](https://doi.org/10.1190/1.1443099).
- Vidale, J., 1988, Finite-difference calculation of travel times: *Bulletin of the Seismological Society of America*, **78**, 2062–2076.
- Vozoff, K., and D. L. B. Jupp, 1975, Joint inversion of geophysical data: *Geophysical Journal of the Royal Astronomical Society*, **42**, 977–991, doi: [10.1111/j.1365-246X.1975.tb06462.x](https://doi.org/10.1111/j.1365-246X.1975.tb06462.x).
- Ward, S. H., and G. W. Hohmann, 1988, Electromagnetic theory for geophysical applications, in M. N. Nabighian, ed., *Electromagnetic methods in applied geophysics*: SEG 1, 131–311.
- Wiggins, R. A., K. L. Lerner, and R. D. Wisecup, 1976, Residual statics analysis as a general linear inverse problem: *Geophysics*, **41**, 922–938, doi: [10.1190/1.1440672](https://doi.org/10.1190/1.1440672).
- Xia, J., R. Miller, and C. B. Park, 1999, Estimation of near-surface shear-wave velocity by inversion of Rayleigh waves: *Geophysics*, **64**, 691–700, doi: [10.1190/1.1444578](https://doi.org/10.1190/1.1444578).
- Yilmaz, O., 2001, *Seismic data analysis: Processing, inversion and interpretation of seismic data*, 2nd ed.: SEG Investigations in Geophysics 10.
- Zhang, J., 1997, Nonlinear refraction and reflection traveltime tomography: Ph.D. thesis, Massachusetts Institute of Technology.
- Zhang, J., 2006, Refraction migration: Imaging multiple refractors automatically: 76th Annual International Meeting, SEG, Expanded Abstracts, 2426.
- Zhang, J., 2009, 12.571 Near-surface geophysical imaging, Fall 2009: Massachusetts Institute of Technology: MIT OpenCourseWare, <http://ocw.mit.edu>, accessed 28 August 2016.
- Zhang, J., and F. D. Morgan, 1996, Joint seismic and electrical tomography: Presented at the EEGS Symposium on Applications of Geophysics to engineering and environmental problems, Environmental and Engineering Geophysics Society.
- Zhang, J., and M. N. Toksöz, 1998, Nonlinear refraction traveltime tomography: *Geophysics*, **63**, 1726–1737, doi: [10.1190/1.1444468](https://doi.org/10.1190/1.1444468).
- Zhdanov, M. S., 2009, *Geophysical electromagnetic theory and methods*, Methods in Geochemistry and Geophysics: Elsevier 43, 868.
- Zhdanov, M. S., 2015, *Inverse theory and applications in geophysics*, 2nd ed.: Elsevier.
- Zhu, X., P. Valasek, B. Roy, S. Shaw, J. Howell, S. Whitney, N. D. Whitmore, and P. Anno, 2008, Recent applications of turning-ray tomography: *Geophysics*, **73**, no. 5, VE243–VE254, doi: [10.1190/1.2957894](https://doi.org/10.1190/1.2957894).